

Parametric study, master report

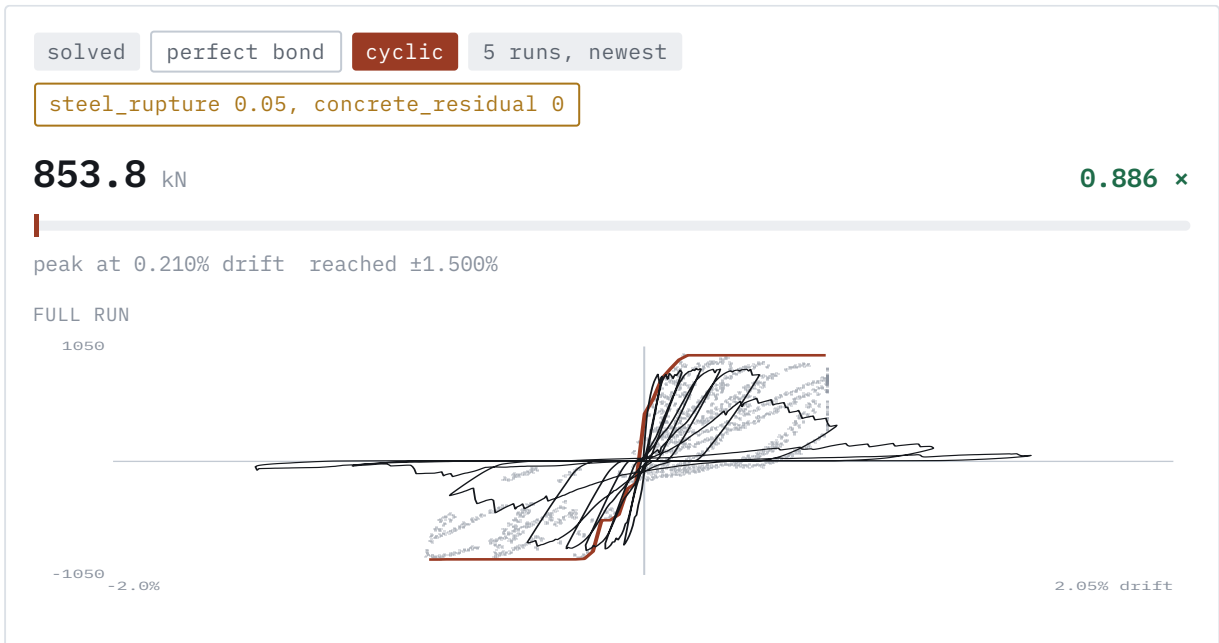
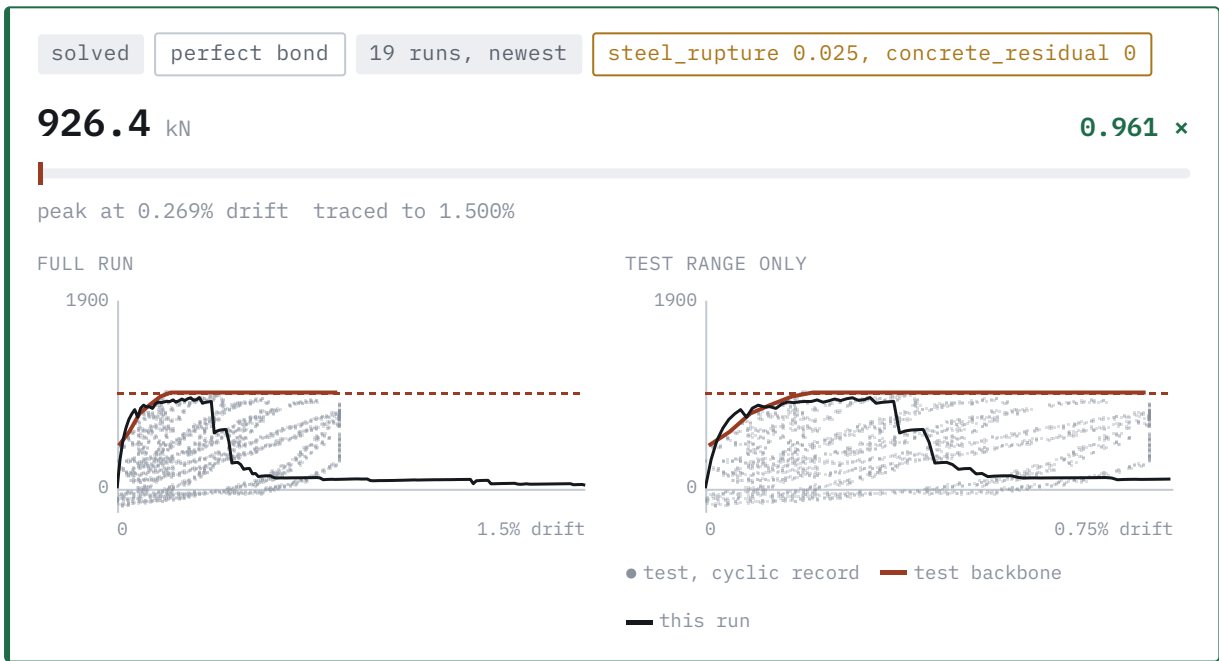
Specimen: 3000 × 2250 × 210 mm squat wall, aspect ratio 0.75, no axial load, tested by **Aldemir, Binici & Canbay (2017)**, ACI SJ 114(2). Lattice method and the simulation being replicated: **Aydin, Tuncay & Binici (2019)**, J. Struct. Eng. 145(9).

45 runs. Generated 2026-09-12 18:06. Cells with no run are gaps, not omissions.

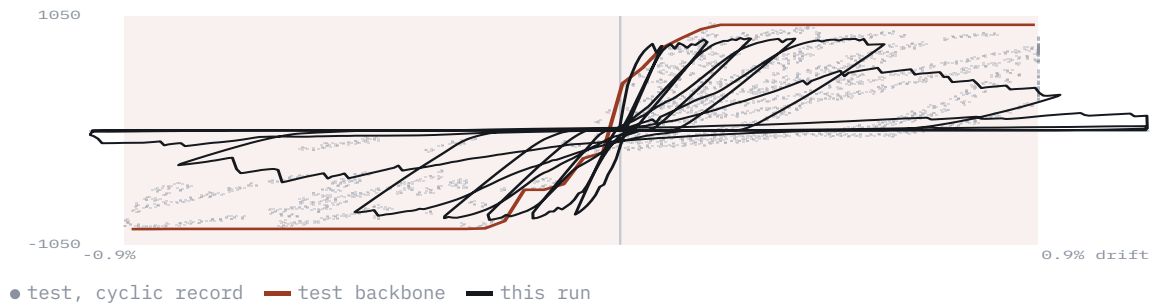
963.6 kN measured peak — Aldemir, Binici & Canbay (2017), Table 4. Every ratio below is against this.

THE MATRIX — ONE ROW PER CELL PER ANALYSIS

crushing Concrete02: yields, softens, crushes



TEST RANGE ONLY



solved

bond60

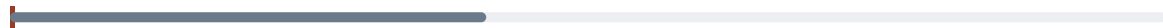
2 runs, newest

steel_rupture 0.05, concrete_residual 0

START-UP SPIKE

1,321.5 kN

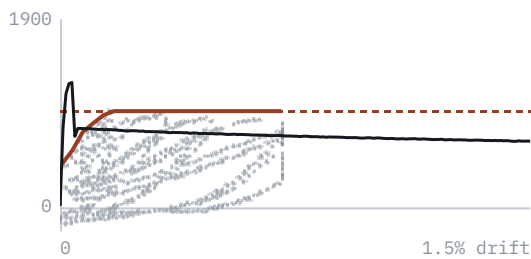
1.371 x



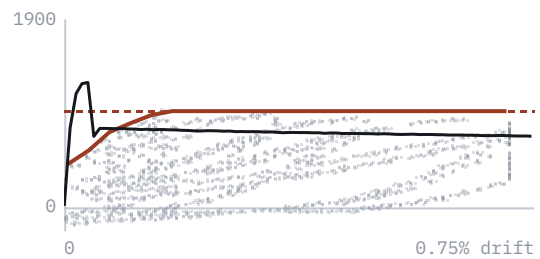
peak at 0.045% drift traced to 1.500%

Sustained past 0.1% drift: **793.7 → 665 kN** — what it actually holds.

FULL RUN



TEST RANGE ONLY



● test, cyclic record — test backbone
— this run

solved

bond70

2 runs, newest

bond damage

START-UP SPIKE

1,775.3 kN

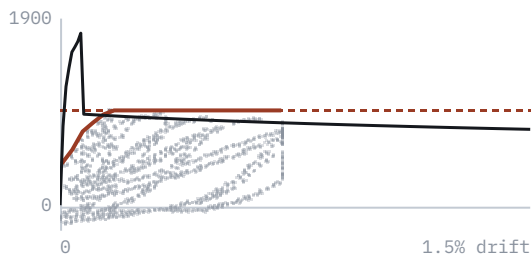
1.842 x



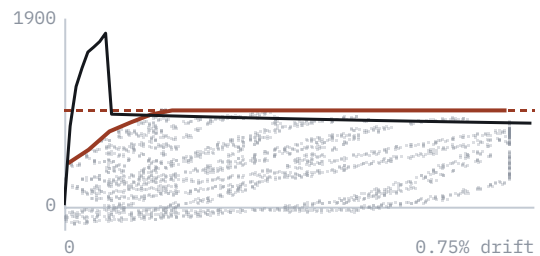
peak at 0.068% drift traced to 1.500%

Sustained past 0.1% drift: **926.6 → 777.6 kN** — what it actually holds.

FULL RUN



TEST RANGE ONLY



● test, cyclic record — test backbone
— this run

capped elastic-perfectly-plastic: yields at f_c , cannot crush

solved

perfect bond

static

268.9 kN

0.279 x

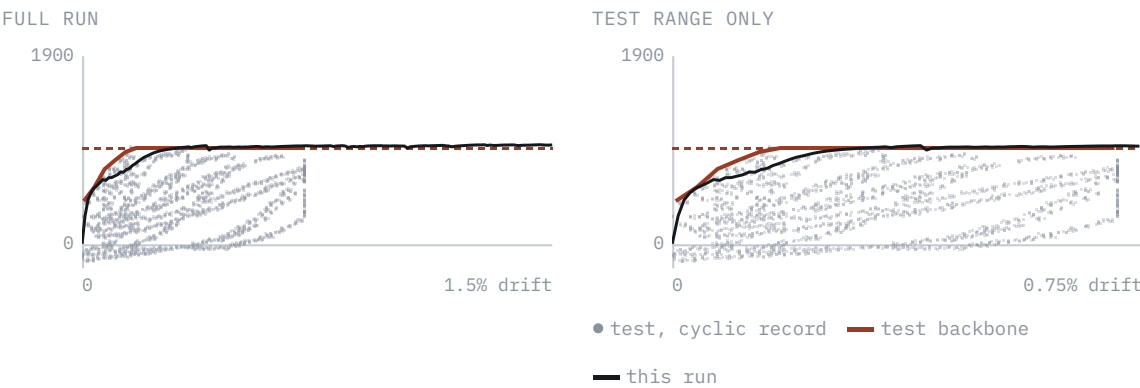
peak at 0.010% drift traced to 0.010% raw maximum, not re-scored

solved perfect bond 3 runs, newest **steel_rupture 0.05**

1,005.6 kN

1.044 x

peak at 1.288% drift traced to 1.500%



paper perfect bond **static**

321.3 kN

0.333 x

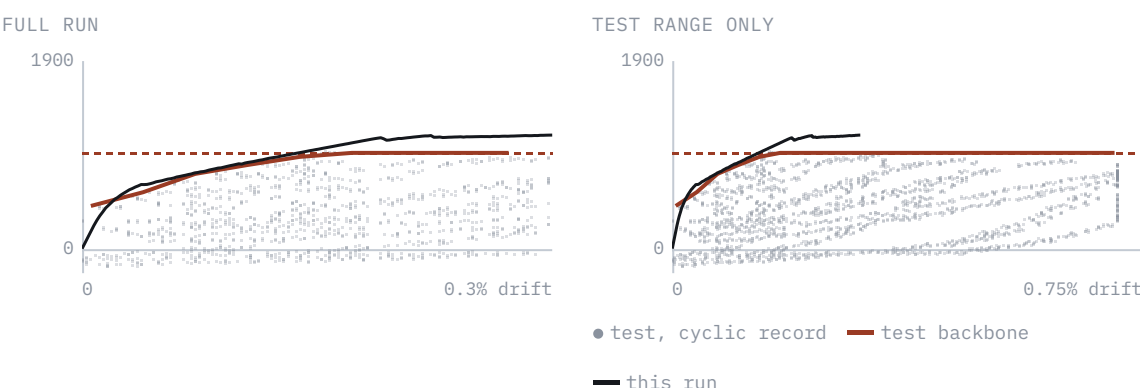
peak at 0.012% drift traced to 0.012% raw maximum, not re-scored

paper perfect bond

1,149.8 kN

1.193 x

peak at 0.300% drift traced to 0.300%



linear Aydin's own law: compression never yields

solved perfect bond **static**

268.9 kN

0.279 x

peak at 0.010% drift traced to 0.010% raw maximum, not re-scored

solved

perfect bond

3 runs, newest

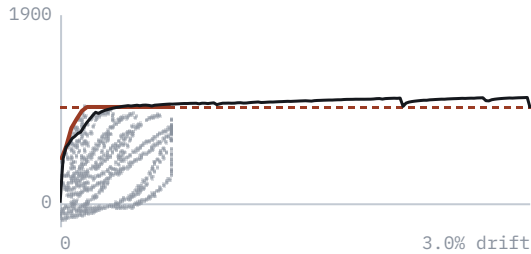
steel_rupture 0.05

1,079.3 kN

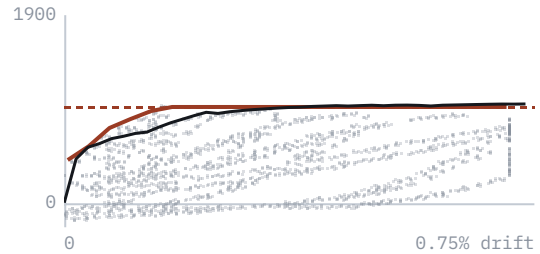
1.120 ×

peak at 2.989% drift traced to 3.000%

FULL RUN



TEST RANGE ONLY



● test, cyclic record — test backbone

— this run

solved

bond60

START-UP SPIKE

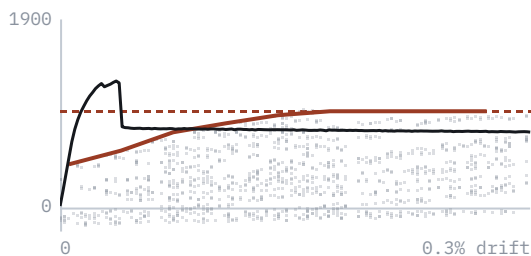
1,286.0 kN

1.335 ×

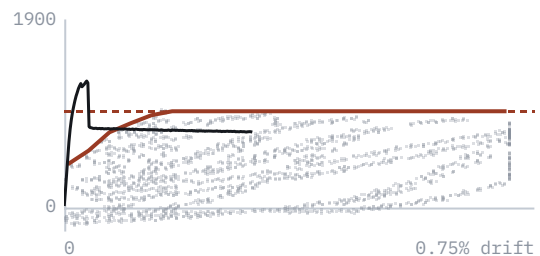
peak at 0.036% drift traced to 0.300%

Sustained past 0.1% drift: **790.7 → 758.1 kN** — what it actually holds.

FULL RUN



TEST RANGE ONLY



● test, cyclic record — test backbone

— this run

paper

perfect bond

static

321.3 kN

0.333 ×

peak at 0.012% drift traced to 0.012% raw maximum, not re-scored

paper

perfect bond

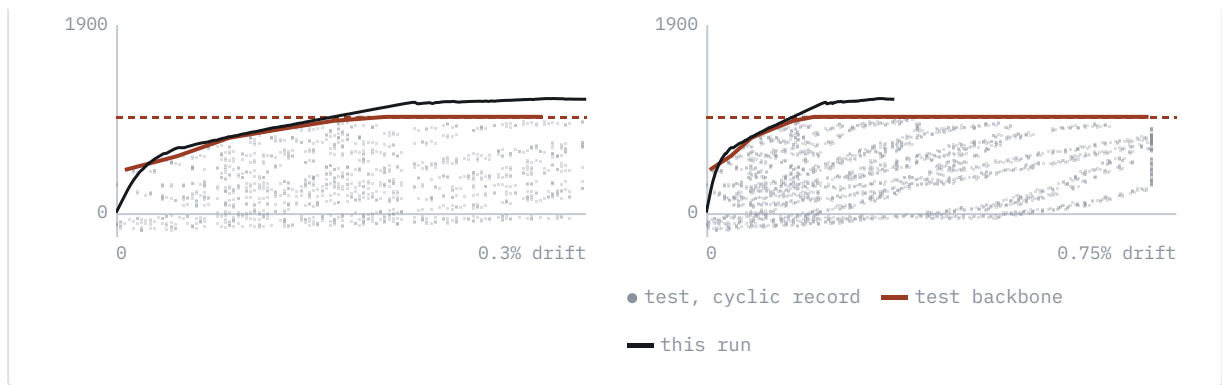
1,155.5 kN

1.199 ×

peak at 0.284% drift traced to 0.300%

FULL RUN

TEST RANGE ONLY



THE TENSION TAIL – WHAT *SOLVED* AND *PAPER* MEAN

Aydin's concrete cracks along a **trilinear tension tail**: linear to (ϵ_{cr}, f_t) , then down through knees at $a_1\epsilon_{cr}$ and $a_2\epsilon_{cr}$, reaching zero at $a_3\epsilon_{cr}$. The shape constants a_1, b_1, b_2 are the same for every specimen in his Table 1. **How far the tail runs — a_2 and a_3 — is the axis.**

solved — a_2 and a_3 are computed per strut so it dissipates exactly $G_f = 75 \text{ N/m}$ over its own length. A longer strut gets a shorter tail. That is what makes the model **mesh-objective**.

paper — $a_2 = 70, a_3 = 360$, the values he prints for this specimen, fitted at **his** grid size of 20 mm. This study runs at 50 mm, and transplanting a fitted tail across grid sizes breaks the energy identity: measured, it dissipates **$5.26 \times G_f$** . It buys +22.6% base shear, which is why the paper cells overshoot at $1.19 \times$ and why the column is **dropped rather than pursued**.

The axis exists only for the two laws built on his trilinear backbone. **crushing** uses Concrete02, whose tail is bilinear and regularised differently, so **crushing / paper cannot be built** — three of the gaps are permanent.

READING THE BARS AND THE CHARTS

Bar length is the overshoot above the measured 963.6 kN; the oxide tick at the left edge is the measurement itself.

Each analysis carries two panes of the same data — the **full run** to its own reach, and the **test range only**, cut where the record stops. Both use one vertical scale, so the dashed oxide line at 963.6 kN sits at a single height throughout and the panes differ only in x.

The matrix is keyed on compression law $\times$ tension tail $\times$ bond, so **any other parameter shadows**: the newest run in a cell may be a mesh or damping variant standing where the baseline was. Each row therefore names what is non-default about the run actually shown — an amber chip such as **mesh 25** means you are not looking at the cell's plain baseline.

Cells marked **start-up spike** peak below 0.1% drift, where a bond ring acts as a parallel load path. That force is real but it is not sustained resistance; each such row prints what it actually holds afterwards.

ANALYSES PRESENT

32	5	4	4
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PUSHOVER	CYCLIC	ELASTIC	STATIC
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Two runs answer the same question when every model- and analysis-affecting parameter matches except the axis under study.

Data as of 2026-09-12 18:06. Rebuilt from the run directories and republished when the study advances — this page carries its own copy rather than fetching one.